

The Arborist

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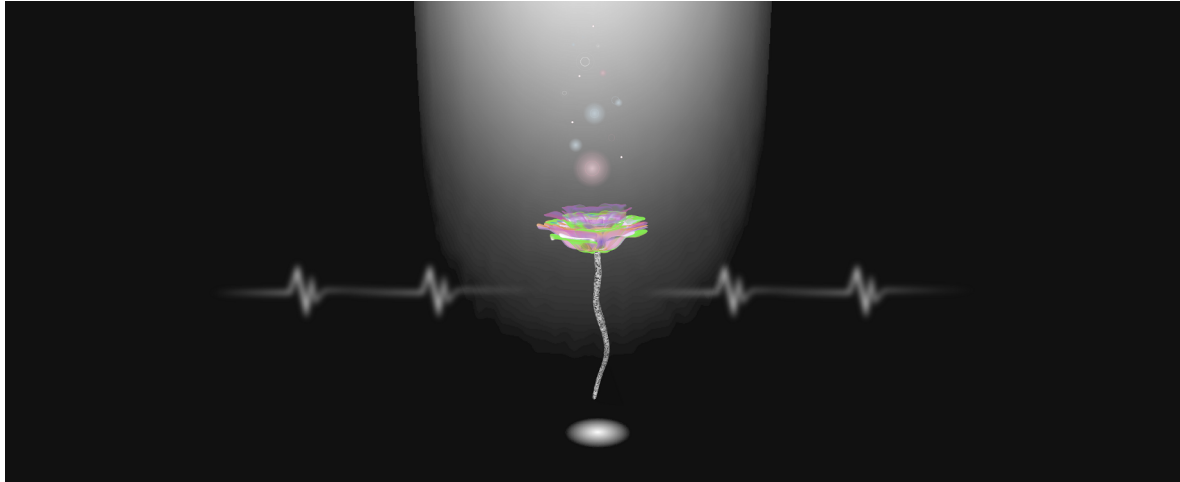


Figure 1: Conceptual overview of the MR experience, where bio-signals power a flower.

ABSTRACT

The Arborist is a mixed reality (MR) installation transforms physiological signals into digital blooms, inviting participants to co-create a shared virtual ecosystem through embodied interaction. Rather than storing raw bio-signals, the system maps heart rate, skin conductance, and temperature to generative visual and sonic parameters, rendering internal states as ambient, non-indexical forms. Users grow and place digital flowers on a tree, transitioning between MR and virtual reality (VR) through gesture. The evolving visual archive preserves only aesthetic traces of presence. Rooted in posthuman and participatory design, the work reimagines biometric data as poetic, collaborative memory connecting past and future.

Index Terms: Mixed Reality (MR), Biometrics, Participatory Art, Shader.

1 INTRODUCTION

Biometric and physiological signals are culturally charged, functioning as tools of measurement, surveillance, and identity [1]. From hospital monitors and airport scanners to fitness trackers and lie detectors, they position the body within systems of verification and control. Although deeply personal, these signals are rarely owned by the individuals who generate them, often symbolizing vulnerability and objectification.

It is through aesthetic and technological mediation that biometrics acquire meaning. Works such as the photosensor-based *I Light U Up* [4] or the EEG-driven *SyncoCreate* [2] transform intimate

bodily data into expressive visual forms. These artistic translations render physiological activity interpretable, shaping shared imaginaries of health, identity, and future human-machine relations.

2 CONCEPT AND ARTISTIC VISION

We reframe biometric data as a medium for collective authorship rather than surveillance. The installation explores how physiological signals - pulse, breath, temperature - can seed a digital landscape that grows with each user's presence. Drawing on naturalistic metaphors for renewal and communal cultivation, the piece envisions data as nourishment rather than extraction. Each gesture of participation contributes to an ever-growing digital ecology that links individual embodiment with collective, temporal memory that links the body, the environment, and technology.

3 USER EXPERIENCE

The user journey is illustrated in Figure 2. The experience begins in VR and shifts to MR, guiding the user to focus on a floating pad (a). The user is invited to touch a glowing pad (b) that links the physical and virtual worlds. A white flower appears in an uncolored form awaiting activation (c). As the user holds it, the bloom absorbs their physiological signals, gradually gaining color and motion (d).

With a double-pinch gesture, the flower compresses into a data sphere which holds the embodied input of user's physiological data (e). When placed onto the shared tree, it grows a new cluster of leaves (f). Each participant's trace remains visible as others join, forming an ever-evolving archive of collective presence.

4 TECHNICAL IMPLEMENTATION

4.1 Real-Time Shader Visuals

Procedurally generated fragments respond to bio-signals in real time via custom shaders. Heart rate (HR) drives a soft, breath-like motion cascading across the digital mesh, mapping physiological rhythms to continuous visual change.

This effect is achieved by layering multiple Fresnel shaders with varied threshold sensitivities (Figure 3). The Fresnel accentuates

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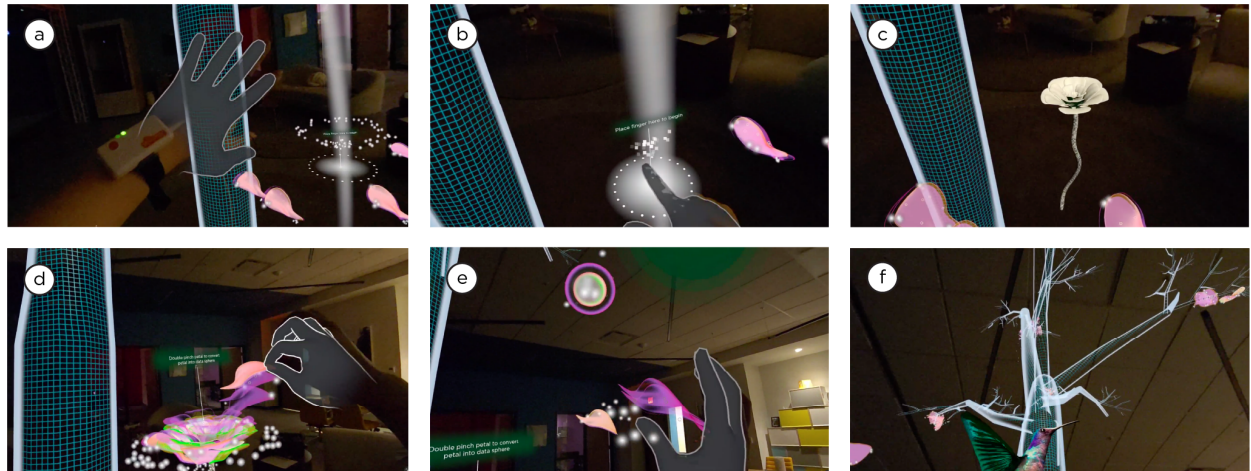


Figure 2: User journey sequence: from initiating touch and generating a bloom to contributing to a shared digital tree.

the organic curvature of the floral mesh, creating a subtle glow that pulses with HR fluctuations. A dynamic noise texture simulates drifting particles such as pollen, while blending multiple materials with varying thresholds adds depth and motion.

Shader parameters update in real time based on the sensor input: skin temperature defines the base hue of the petals; HR modulates Fresnel intensity and particle speed; and galvanic skin response (GSR) controls particle size and secondary color values.

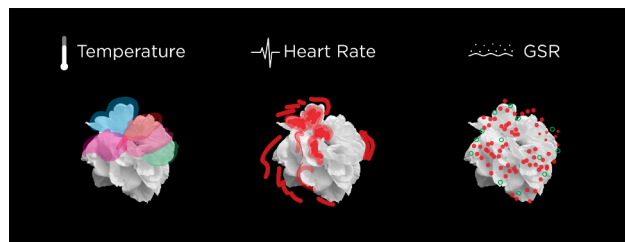


Figure 3: Real-time shader art driven by sensor input.

4.2 The VR/MR Transition

The system allows users to transition between MR and VR, enabling a deeper connection between their physical presence and digital expression. This real-time transition functions as both an interaction mechanic and a metaphor, representing the flow of physiological data from the physical world into a virtual space. In MR mode, real-world awareness is maintained through passthrough rendering, while VR mode presents a fully immersive environment via a dynamic skybox. The transition is gesture-based: when the user cups their hands, the system detects the gesture and initiates the transition into VR. When the hands separate, the environment seamlessly reverts to MR.

4.3 The Adaptive Sound

The soundscape for the experience is based on the framework used for interactive audio in video games. The musical score is comprised of several simultaneous looping audio tracks, each characterized by a single virtual musical instrument. Sound effects triggered at runtime are dynamically mixed into the soundtrack such that they respect both the harmonic structure of the music and the relatively constant volume of the soundscape. Physiological data are then employed as effects controllers along the musical signal chain. Users

influence reverb, filters, and mixing levels with their own bodies, acting both as listeners and composers. Thus, although the musical material is pre-composed, embodied actions shape the sonic texture in meaningful ways. The soundtrack was composed using Ableton Live 10, with interactive logic implemented in FMOD for Unity.

4.4 The Accumulated Memory

Rather than storing raw biometric data, our system records only visual parameters abstracted from each user's physiological signals. Building on prior work in save-and-load systems for MR applications (reference omitted for anonymization) [3], a lightweight local storage mechanism preserves these impressions. During each session, signals are normalized and mapped in real time to attributes such as color, scale, and animation, generating a unique digital bloom saved as a compact JSON file. Over time, these non-identifiable records accumulate into a layered archive shaped by the collective contributions of past participants.

5 REFLECTION AND CONTRIBUTION

The Arborist treats biometric computing as communion rather than classification. It invites people to consider how their internal states can nurture a digital ecosystem shared with others. By allowing past contributions to shape the environment encountered by future viewers, the work reflects on continuity, collective memory, and the evolving entanglement of bodily presence and digital worlds.

ACKNOWLEDGMENTS

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